

NBME-Style Questions: Hyponatremia

QUESTION BANK

Question 1

A 68-year-old woman is found to have a serum sodium of 128 mEq/L on routine laboratory work. Which of the following serum sodium values defines hyponatremia?

- A) < 145 mEq/L
 - B) < 140 mEq/L
 - C) < 135 mEq/L
 - D) < 130 mEq/L
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Question 2

A 55-year-old man with a history of chronic diarrhea presents with fatigue and dizziness. Serum sodium is 128 mEq/L. Physical examination reveals dry mucous membranes, decreased skin turgor, and orthostatic hypotension. Urine sodium is 12 mEq/L. Which of the following is the most appropriate first-line treatment?

- A) Fluid restriction
 - B) Hypertonic saline (3% NaCl)
 - C) Isotonic saline (0.9% NaCl)
 - D) Furosemide and fluid restriction
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Question 3

A 72-year-old man with congestive heart failure presents with bilateral lower extremity edema and a serum sodium of 126 mEq/L. Which of the following best describes the pathophysiology of his hyponatremia?

- A) Low total body sodium with proportionally greater water loss
 - B) Normal total body sodium with excess free water retention
 - C) Excess total body sodium with proportionally greater water retention
 - D) Excess total body sodium with proportionally greater sodium retention
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Question 4

A 45-year-old woman with small cell lung cancer is found to have a serum sodium of 122 mEq/L. She appears euvolemic on examination. Urine osmolality is 450 mOsm/kg and urine sodium is 55 mEq/L. Which of the following is the most likely diagnosis?

- A) Psychogenic polydipsia
 - B) Hypothyroidism
 - C) Syndrome of inappropriate antidiuretic hormone secretion (SIADH)
 - D) Beer potomania
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Question 5

A patient with SIADH is being treated with hypertonic saline. His sodium corrects from 118 mEq/L to 130 mEq/L within 12 hours. Two days later, he develops dysarthria, difficulty swallowing, and flaccid paralysis of his lower extremities. Which of the following is the most likely diagnosis?

- A) Cerebral edema
 - B) Central pontine myelinolysis (Osmotic Demyelination Syndrome)
 - C) Wernicke's encephalopathy
 - D) Hypertensive encephalopathy
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Question 6

A 38-year-old woman with psychiatric illness presents with nausea and confusion. Her serum sodium is 124 mEq/L. She is euvolemic on examination. Urine osmolality is 60 mOsm/kg and urine sodium is 15 mEq/L. Which of the following is the most likely cause of her hyponatremia?

- A) SIADH
 - B) Hypothyroidism
 - C) Psychogenic polydipsia
 - D) Addison's disease
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Question 7

A 60-year-old man with cirrhosis presents with increasing abdominal girth and bilateral leg swelling. Serum sodium is 129 mEq/L. Which of the following is the most appropriate first-line treatment for his hyponatremia?

- A) Isotonic saline infusion
 - B) Hypertonic saline infusion
 - C) Fluid restriction and furosemide
 - D) Tolvaptan monotherapy
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Question 8

A physician plans to correct a patient's sodium from 115 mEq/L to a safe level over 24 hours. According to safe correction guidelines, what is the maximum rate of sodium correction per 24 hours to avoid osmotic demyelination syndrome?

- A) 2–4 mEq/L per 24 hours
 - B) 6–8 mEq/L per 24 hours
 - C) 10–12 mEq/L per 24 hours
 - D) 15–20 mEq/L per 24 hours
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Question 9

A 50-year-old man presents with hyponatremia, hypotension, hyperkalemia, and hyperpigmentation of the skin. Which of the following is the most likely underlying cause of his hyponatremia?

- A) SIADH
 - B) Congestive heart failure
 - C) Addison's disease
 - D) Nephrotic syndrome
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Question 10

A 33-year-old woman presents with hyponatremia. Urine sodium is 8 mEq/L and she has signs of volume depletion. Which of the following best explains the low urine sodium in this patient?

- A) Renal sodium wasting due to diuretic use
 - B) Extrarenal sodium loss with appropriate renal sodium conservation
 - C) Impaired ADH secretion
 - D) Excess aldosterone production causing urinary sodium retention
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Question 11

A 29-year-old man who drinks 12 beers per day and eats very little presents with confusion and a serum sodium of 125 mEq/L. His ADH levels are normal. Which of the following best explains his hyponatremia?

- A) Excess ADH secretion from ectopic tumor
 - B) Adrenal insufficiency causing sodium wasting
 - C) Low solute intake impairing free water excretion
 - D) Psychogenic polydipsia causing dilutional hyponatremia
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Question 12

A 66-year-old woman is started on a medication for recurrent SIADH due to a central nervous system disorder. The drug works by blocking the effect of ADH at the collecting duct. Which of the following medications is being described?

- A) Tolvaptan
 - B) Furosemide
 - C) Demeclocycline
 - D) Conivaptan
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Question 13

A 58-year-old man with nephrotic syndrome develops hyponatremia with a serum sodium of 130 mEq/L. He has significant peripheral edema and ascites. Which volume state best characterizes this patient's hyponatremia?

- A) Hypovolemic hyponatremia
 - B) Euvolemic hyponatremia
 - C) Hypervolemic hyponatremia
 - D) Pseudohyponatremia
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Question 14

A patient with known SIADH has a urine osmolality of 520 mOsm/kg and urine sodium of 60 mEq/L. Which of the following mechanisms best explains the persistently elevated urine osmolality in SIADH?

- A) Increased aldosterone secretion promoting water retention
 - B) Inappropriate continued secretion of ADH causing water retention in the collecting duct
 - C) Decreased GFR leading to concentrated urine
 - D) Activation of the renin-angiotensin system
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Question 15

A 44-year-old woman with hyponatremia secondary to hypothyroidism is diagnosed. Which volume state is most commonly associated with hypothyroid-related hyponatremia?

- A) Hypovolemic
 - B) Euvolemic
 - C) Hypervolemic
 - D) Pseudohyponatremic
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Question 16

A 70-year-old man with CHF and a serum sodium of 127 mEq/L is started on tolvaptan. Which of the following best describes the mechanism of action of tolvaptan?

- A) Inhibits sodium reabsorption in the loop of Henle
 - B) Blocks V2 receptors in the collecting duct, causing aquaresis
 - C) Blocks ADH synthesis in the hypothalamus
 - D) Promotes aldosterone secretion to retain sodium
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Question 17

A 52-year-old man with vomiting for 5 days presents with a serum sodium of 126 mEq/L. His urine sodium is 35 mEq/L. Which of the following best explains the elevated urine sodium despite hypovolemia?

- A) Psychogenic polydipsia
 - B) SIADH
 - C) Diuretic use causing renal sodium wasting
 - D) Adrenal insufficiency
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Question 18

A 61-year-old woman has a serum sodium of 124 mEq/L. She is euvolemic. Urine osmolality is 280 mOsm/kg and urine sodium is 18 mEq/L. Which of the following is the most likely diagnosis?

- A) SIADH
 - B) Psychogenic polydipsia
 - C) CHF-related hyponatremia
 - D) Addison's disease
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Question 19

A patient develops osmotic demyelination syndrome after overly rapid correction of hyponatremia. Which of the following symptom triads is classically associated with this condition?

- A) Confusion, seizures, and papilledema
 - B) Dysarthria, dysphagia, and flaccid paralysis
 - C) Fever, neck stiffness, and photophobia
 - D) Tremor, bradykinesia, and rigidity
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Question 20

A 48-year-old woman with a history of diuretic use presents with hyponatremia. Her urine sodium is 45 mEq/L. Physical examination reveals dry mucous membranes and poor skin turgor. Which of the following volume states and urine sodium pattern best characterizes her condition?

- A) Hypovolemic; urine $\text{Na}^+ > 20$ consistent with renal loss
 - B) Hypovolemic; urine $\text{Na}^+ < 20$ consistent with extrarenal loss
 - C) Euvolemic; urine $\text{Na}^+ > 40$ consistent with SIADH
 - D) Hypervolemic; urine $\text{Na}^+ > 20$ consistent with renal failure
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Question 21

A 30-year-old woman with no significant past medical history presents with hyponatremia and is found to have a urine osmolality of 85 mOsm/kg. She admits to drinking approximately 10 liters of water per day because she believes it is healthy. Which of the following is the most likely diagnosis?

- A) SIADH
 - B) Beer potomania
 - C) Psychogenic polydipsia
 - D) Hypothyroidism
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Question 22

A 65-year-old man with cirrhosis and ascites has a serum sodium of 128 mEq/L. His physician wants to use a vaptane to help correct the sodium. Which of the following receptor is targeted by this class of drugs?

- A) V_1 receptor
 - B) V_2 receptor
 - C) Aldosterone receptor
 - D) AT_1 receptor
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Question 23

A 55-year-old woman with small cell carcinoma of the lung develops hyponatremia. Labs show urine osmolality of 600 mOsm/kg and urine sodium of 70 mEq/L. Serum osmolality is low. Which of the following is the most appropriate first-line treatment for mild-to-moderate symptoms?

- A) Isotonic saline
 - B) Fluid restriction
 - C) Furosemide alone
 - D) Hypertonic saline
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Question 24

A 42-year-old man is found to have hyponatremia with a serum sodium of 120 mEq/L. He has active seizures. Which of the following is the most appropriate immediate treatment?

- A) Fluid restriction
 - B) Oral sodium chloride tablets
 - C) Hypertonic saline (3% NaCl)
 - D) Isotonic saline with potassium supplementation
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Question 25

A 77-year-old woman with heart failure, cirrhosis, and nephrotic syndrome all cause hyponatremia through a shared mechanism. Which of the following best describes this common pathophysiologic mechanism?

- A) Direct renal tubular sodium wasting
- B) Excess ADH-independent free water excretion
- C) Effective arterial blood volume depletion triggering ADH release and water retention
- D) Primary hyperaldosteronism causing dilutional hyponatremia

ANSWER KEY

Question	Correct Answer	Explanation
1	C) < 135 mEq/L	Hyponatremia is defined as serum Na ⁺ < 135 mEq/L per the lecture definition.
2	C) Isotonic saline (0.9% NaCl)	Hypovolemic hyponatremia from extrarenal loss (urine Na ⁺ < 20) is treated with isotonic saline to restore volume.
3	C) Excess total body sodium with proportionally greater water retention	Hypervolemic hyponatremia (CHF) involves excess Na ⁺ AND water, but water is retained in greater proportion.
4	C) SIADH	Classic SIADH criteria: euvolemia, urine osm > 100 (here 450), urine Na ⁺ > 40 (here 55), and an associated cause (small cell lung cancer).
5	B) Osmotic Demyelination Syndrome	Overly rapid correction of hyponatremia (>6–8 mEq/L per 24h) causes ODS, presenting with dysarthria, dysphagia, and flaccid paralysis.
6	C) Psychogenic polydipsia	Very dilute urine (osm 60, low urine Na ⁺) in a euvolemic psychiatric patient indicates excess free water intake overwhelming excretory capacity, not ADH-driven.
7	C) Fluid restriction and furosemide	Hypervolemic hyponatremia (cirrhosis) is treated with fluid restriction plus a loop diuretic (furosemide).
8	B) 6–8 mEq/L per 24 hours	The safe correction rate is no faster than 6–8 mEq/L per 24 hours to prevent osmotic demyelination syndrome.

9	C) Addison's disease	Addison's disease (adrenal insufficiency) causes hyponatremia, hypotension, hyperkalemia, and hyperpigmentation due to loss of aldosterone and cortisol.
10	B) Extrarenal sodium loss with appropriate renal sodium conservation	Urine $\text{Na}^+ < 20$ indicates the kidneys are appropriately conserving sodium, meaning the loss is from an extrarenal source (vomiting, diarrhea).
11	C) Low solute intake impairing free water excretion	Beer potomania: low dietary solute limits the kidney's ability to excrete free water, causing hyponatremia even with normal ADH.
12	C) Demeclocycline	Demeclocycline blocks ADH action at the collecting duct (nephrogenic DI effect); tolvaptan blocks V2 receptors specifically.
13	C) Hypervolemic hyponatremia	Nephrotic syndrome causes excess total body Na^+ and water with relatively more water, resulting in hypervolemic hyponatremia.
14	B) Inappropriate continued ADH secretion causing water retention	In SIADH, ADH continues to act on collecting duct aquaporins, reabsorbing free water and producing concentrated urine despite low serum osmolality.
15	B) Euvolemic	Hypothyroidism causes euvolemic hyponatremia, classified alongside SIADH and psychogenic polydipsia.
16	B) Blocks V2 receptors in the collecting duct, causing aquaresis	Tolvaptan is a selective V2 receptor antagonist (vaptane) that blocks ADH-mediated water reabsorption, increasing free water excretion (aquaresis).
17	C) Diuretic use causing renal sodium wasting	In hypovolemic hyponatremia, urine $\text{Na}^+ > 20$ points to a renal cause; diuretics are listed causes of renal sodium loss in the lecture.
18	B) Psychogenic polydipsia	Euvolemic hyponatremia with dilute urine (osm 280, near isosthenuria) and low urine Na^+ points to excess water intake rather than ADH excess (SIADH would show concentrated urine).

19	B) Dysarthria, dysphagia, and flaccid paralysis	The classic triad of ODS per the lecture is dysarthria, dysphagia, and flaccid paralysis.
20	A) Hypovolemic; urine Na⁺ > 20 consistent with renal loss	Diuretics cause renal sodium wasting (urine Na ⁺ > 20) with resultant hypovolemia — this is a renal cause of hypovolemic hyponatremia.
21	C) Psychogenic polydipsia	Very dilute urine (osm 85 mOsm/kg) with massive water intake in a young, otherwise healthy woman is classic for psychogenic polydipsia.
22	B) V₂ receptor	Vaptans (tolvaptan, conivaptan) are V ₂ receptor antagonists. V ₂ receptors mediate ADH's water-reabsorbing effect in the collecting duct.
23	B) Fluid restriction	SIADH (from small cell lung cancer) with mild-to-moderate symptoms is treated first with fluid restriction; hypertonic saline is reserved for severe/symptomatic cases.
24	C) Hypertonic saline (3% NaCl)	Severe symptomatic hyponatremia (active seizures) requires urgent treatment with hypertonic saline to acutely raise sodium and stop seizure activity.
25	C) Effective arterial blood volume depletion triggering ADH release and water retention	CHF, cirrhosis, and nephrotic syndrome all reduce effective circulating volume, stimulating baroreceptors and causing non-osmotic ADH release with subsequent water retention.

Study Tip: For boards, master the volume status → urine Na⁺ → treatment algorithm. Always remember: correct Na⁺ no faster than **6–8 mEq/L per 24 hours** to avoid ODS. Know the SIADH diagnostic criteria: urine osm > 100, urine Na⁺ > 40, euvolemic state.